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A team of investigators determined the mean pulmonary artery pressure in individuals with a large range of body mass indexes who were undergoing a cardiac procedure and classified these individuals according to standard definitions of obesity. These definitions are based on an individual's body mass index (BMI).

Table 1 Mean Pulmonary Artery Pressure by Obesity Status

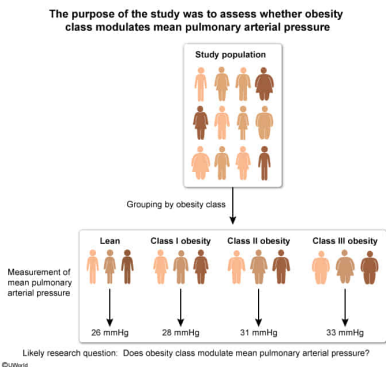
Group	Mean pulmonary artery pressure (mm Hg)
Lean (average BMI 22)	26
Class I obese (average BMI 32)	28
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The study described in the passage was conducted for the purpose of:

- ☐ A. determining whether the presence or absence of obesity affects mean pulmonary artery pressure. [17%]
- ☐ B. determining whether mean pulmonary artery pressure affects BMI. [1%]
- ☒ C. determining whether obesity class affects mean pulmonary artery pressure. [80%]
- ☐ D. determining whether pulmonary systolic and diastolic pressures affect obesity levels.

Correct
79% answered correctly

Explanation:



Blood pressure refers to the pressure exerted against blood vessel walls by the blood they contain. The arterial pressures during systole or diastole are referred to as **systolic** and **diastolic pressures**, respectively, and can be used along with the relative durations of systole and diastole to estimate the **mean arterial pressure** (MAP).

The passage describes a study in which a sample of individuals possessing a wide range of body compositions was studied. The passage indicates that these individuals were divided into groups according to defined obesity classes, and Table 1 presents the mean pulmonary artery pressure in each of the groups. Because the **investigators grouped** the study **participants according to obesity** class and the **mean arterial pressure** data are **presented separately for each group**, the **likely purpose of the study** was to **determine whether obesity class affects mean pulmonary artery pressure**.

(Choice A) Because the investigators classified the subjects into distinct classes and reported the results for each class, it is likely they wished to study the relationship between obesity and pulmonary artery pressure in finer detail than would be found in a simple comparison of lean versus obese individuals.

(Choice B) The investigators measured mean pulmonary artery pressure in individuals who had been classified into one of several groups based on BMI. Therefore, mean pulmonary artery pressure was a dependent variable rather than an independent variable, and the purpose of the study was *not* to determine the direct relationship between BMI and mean pulmonary artery pressure.

(Choice D) Although pulmonary systolic and diastolic pressures determine the mean pulmonary artery pressure, pulmonary systolic and diastolic pressures were not reported in the passage. Therefore, it is unlikely the investigators were focused on the individual influences of these measures.

Educational objective:

Blood pressure refers to the pressure exerted against blood vessel walls during heart contraction (systole) and relaxation (diastole).

Time spent:0 sec
Subject:
Biology

QID:403779
System:
Circulation and Respiration

Approximately 40% of adults and 20% of children in the United States are obese. Obese individuals whose excess adipose tissue is mainly found in the abdomen and thorax are sometimes referred to as displaying abdominal or central obesity, whereas those whose excess adipose tissue is located more along the hips, thighs, and buttocks are said to display peripheral obesity.

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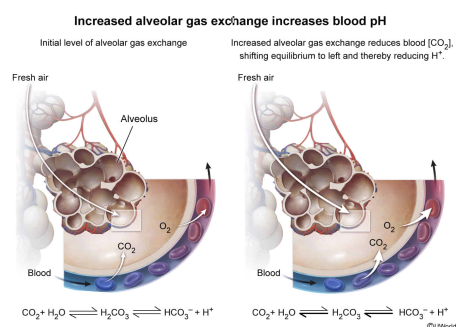
What condition would tend to increase blood pH in an obese individual at rest?

- ☐ A. An increase in blood CO₂ levels [24%]
- ☐ B. A decrease in breathing rate [21%]
- ☒ C. An increase in the volume of fresh air reaching the alveoli [41%]
- ☐ D. A decline in oxygen uptake by red blood cells [12%]

Correct

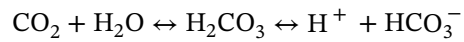
40% answered correctly

Explanation:



The body maintains blood pH within a relatively narrow range to avoid harmful effects due to excess acidity or basicity. Similarly, adequate O₂ levels and levels of CO₂ consistent with homeostasis in the blood are required for

human life. Various **buffer** systems contribute to **pH regulation** of the blood and tissues, and one of these, the **bicarbonate buffer system**, links changes in blood CO₂ and H⁺:



An enzyme called carbonic anhydrase in red blood cells catalyzes the interconversion of CO₂ and water with carbonic acid (H₂CO₃), thereby allowing relatively rapid equilibration to occur when blood CO₂ or pH levels change.

The question asks for the condition that would tend to increase blood pH in a resting obese individual. Due to the **bicarbonate equilibrium** and **Le Chatelier's principle**, removal of CO₂ from the blood will drive increased utilization of H⁺ and HCO₃⁻ (bicarbonate anion) for carbonic acid synthesis. This causes a reduction in blood H⁺ and, correspondingly, an increase in pH. Therefore, an **increase in the volume of air reaching the alveoli and exchanging with the blood** will **increase blood pH** by allowing more CO₂ in the blood to enter the alveoli and be exhaled.

(Choices A and B) A decrease in breathing rate will likely increase the blood CO₂ level. Increased CO₂ levels will in turn drive an increase in H⁺ concentration via the bicarbonate equilibrium. Situations causing an increase in blood CO₂ levels therefore tend to *decrease* the pH (corresponding to an increase in [H⁺]).

(Choice D) A decline in oxygen uptake from the breath corresponds to a reduction in gas exchange across the alveoli into the blood. Under such circumstances, the blood CO₂ level will rise, leading to an increase in [H⁺] and a *reduction* in blood pH rather than an increase in blood pH.

Educational objective:

Changes in respiration affect blood pH levels, with increased respiration tending to cause increased blood pH.

Time spent:0 sec

Subject:
Biology

QID:403780

System:
Circulation and Respiration

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The expiratory reserve volume (ERV) is the volume of air one can forcefully exhale after a normal expiration. Which table shows the expected ERVs of lean and obese individuals, tested both while standing and while lying horizontally in bed?

☐ A.

Group	Condition	Volume (L)
Lean	Standing	0.65
	Lying in bed	0.60
Obese	Standing	2.00
	Lying in bed	1.30

[8%]

☐ B.

Group	Condition	Volume (L)
Lean	Standing	1.30
	Lying in bed	2.00
Obese	Standing	0.60
	Lying in bed	0.65

[8%]

☒ C.

Group	Condition	Volume (L)
Lean	Standing	2.00
	Lying in bed	1.30
Obese	Standing	0.65
	Lying in bed	0.60

[77%]

☐ D.

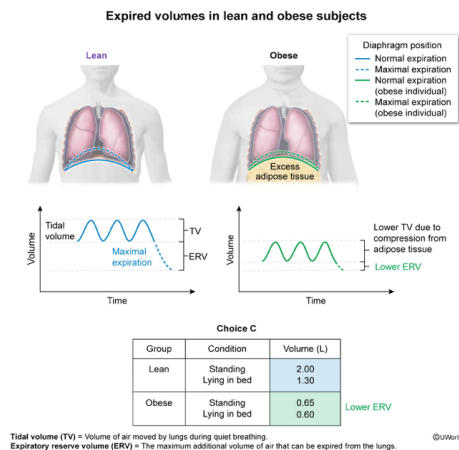
Group	Condition	Volume (L)
Lean	Standing	0.60
	Lying in bed	0.65
Obese	Standing	1.30
	Lying in bed	2.00

[6%]

Correct

76% answered correctly

Explanation:



Ventilation, the movement of air into and out of the lungs, facilitates gas exchange in alveoli. This gas exchange results in transfer of O_2 from the atmosphere to the blood and CO_2 from the blood to the atmosphere. Ventilation also permits the occurrence of important functions such as vocalization and coughing.

Assessment of ventilation is often performed as part of the overall assessment of respiratory function. Such testing is typically carried out by having an individual perform a series of submaximal and maximal expirations and inspirations, thereby allowing determination of specific lung **volumes and capacities**.

The passage states that anatomical changes present in some obese individuals can influence circulatory and respiratory function, and the question asks for the expected ERVs of lean and obese subjects, both while standing and while lying down.

As one goes from a **standing to a reclining position**, **tissues and organs** in the abdomen **shift toward the head**, which can restrict the range of motion of the diaphragm and thereby **reduce the volume of air** that is **expired with maximal effort**. Similarly, excess adipose tissue stored in the thorax and abdomen of individuals with **central obesity exerts a restrictive force on the lungs and diaphragm** that can also **limit maximal expired volume**.

(Choice A) Expiration in lean individuals would be expected to be less (*not more*) restricted (corresponding to a greater volume expired) by abdominal fat than expiration in obese individuals.

(Choice B) Shifting of abdominal organs and tissues would be expected to reduce (*not increase*) the expired volumes of lean and obese individuals during the expiration in bed relative to their expired volumes while standing.

(Choice D) The expired volumes of lean individuals would be expected to be *less* restricted by abdominal fat than the expired volumes of obese individuals, and in both lean and obese individuals the expired volume would be expected to be *greater* while standing than while lying supine in bed.

Educational objective:

Anatomical features (eg, body position, amount of adipose tissue) can influence respiratory and cardiovascular function.

Time spent: 0 sec

Subject:
Biology

QID: 403781

System:
Circulation and Respiration

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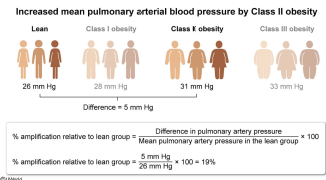
Class II obesity increased mean pulmonary blood pressure relative to the lean group by:

- ☐ A. 7% [12%]
- ☐ B. 11% [12%]
- ☒ C. 19% [69%]
- ☐ D. 27% [5%]

Correct

68% answered correctly

Explanation:



The pumping action of the **heart** is responsible for creating the pressure differences that drive blood flow through the **pulmonary and systemic** circulations. Although the entire cardiac output flows through both circulations, the flow and pressure characteristics in the two circulations are markedly different.

Since the lungs receive as much blood per unit time as all the other body tissues combined, lung blood flow is typically much higher per unit of mass than for the other tissues. Vascular resistance increases with **blood vessel length**, and the length of the pulmonary vasculature is much shorter than that of the systemic vasculature. As a result, the pressure drop along the pulmonary circulation is less than that along the systemic circulation, and high rates of pulmonary blood flow are maintained at much lower pressures than are required in the systemic circulation.

This question asks for the percentage by which mean blood pressure in the pulmonary arteries is increased in the group with Class II obesity relative to the lean group. From Table 1, mean pulmonary artery blood pressure was 26 mm Hg in the lean group and 31 mm Hg in the group displaying Class II obesity. Therefore, the **percent amplification** of mean pulmonary artery blood pressure can be **determined by dividing the difference** in mean **pulmonary artery pressure** (ie, 5 mm Hg) in the two groups **by** the mean pulmonary artery **pressure in the lean group** (26 mm Hg) and **multiplying by 100**:

$$\begin{aligned} \text{\% amplification relative to lean group} &= \frac{\text{Difference in mean pulmonary artery pressure}}{\text{Mean pulmonary artery pressure in the lean group}} \times 100 \\ \text{\% amplification relative to lean group} &= \frac{5 \text{ mm Hg}}{26 \text{ mm Hg}} \times 100 = 19 \% \end{aligned}$$

(Choices A, B, and D) The values in these choices result from incorrect calculation of the percent difference between the lean group and the group displaying Class II obesity.

Educational objective:

The mean arterial blood pressure can be influenced by alterations in anatomy or blood vessel structure that accompany a clinical state (eg, obesity).

Time spent:0 sec

Subject:
Biology

QID:403782

System:
Circulation and Respiration

Approximately 40% of adults and 20% of children in the United States are obese. Obese individuals whose excess adipose tissue is mainly found in the abdomen and thorax are sometimes referred to as displaying abdominal or central obesity, whereas those whose excess adipose tissue is located more along the hips, thighs, and buttocks are said to display peripheral obesity.

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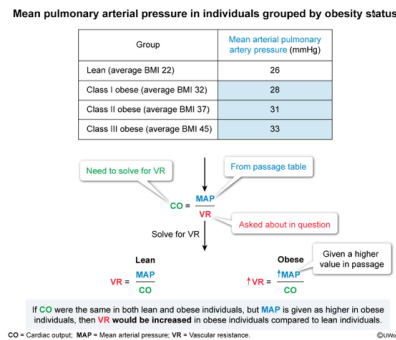
When combined with data in the passage, which finding alone would be most likely to lead the researchers to conclude that pulmonary vascular resistance is increased in the obese subjects?

- ☐ A. Higher heart rates in the obese subjects than in the lean subjects [29%]
- ☒ B. Similar cardiac output in the obese subjects and the lean subjects [9%]
- ☒ C. Decreased stroke volume in the obese subjects relative to that in the lean subjects [41%]
- ☐ D. Upregulation of breathing rate in the obese subjects relative to the lean subjects [20%]

Incorrect. Correct answer is **B**

8% answered correctly

Explanation:



The **cardiac output** is the volume of blood pumped by the heart per unit time. This volume is pumped by the right ventricle to the lungs and by the left ventricle to the rest of the tissues in the body. Cardiac output can be estimated according to the mathematical relationship between blood pressure and resistance along a blood flow circuit:

$$\text{Cardiac output} \approx \frac{\text{Mean arterial pressure}}{\text{Vascular resistance}}$$

Because both the pulmonary and systemic circulations receive the entire cardiac output, cardiac output can be related to both systemic mean arterial blood pressure and systemic vascular resistance as well as to mean pulmonary artery blood pressure and pulmonary vascular resistance.

Table 1 presents the mean pulmonary artery blood pressure in the lean and obese groups of subjects by a team of researchers, and the question asks which additional finding would allow the researchers to conclude that pulmonary vascular resistance is increased in the obese subjects.

Because the pulmonary circulation receives the entire cardiac output, cardiac output can be combined with pulmonary artery blood pressure to draw conclusions about the vascular resistance of the pulmonary circulation:

$$\text{Pulmonary vascular resistance} \approx \frac{\text{Mean pulmonary artery pressure}}{\text{Cardiac output}}$$

According to Table 1, mean pulmonary artery blood pressure was on average higher in all the obese groups than in the lean group. Therefore, a **finding** that **cardiac output** is **similar in** the **obese** and **lean** subjects **would suggest** that **pulmonary vascular resistance** was **increased in** the **obese subjects**.

(Choice A) Although heart rate is a determinant of cardiac output, there are no data in the passage (eg, stroke volume) that would allow the researchers to use heart rate to conclude that pulmonary vascular resistance is higher in the obese subjects.

(Choice C) By itself (ie, in the absence of heart rate data), stroke volume does not allow any conclusions to be drawn about pulmonary vascular resistance when considered along with the data (ie, mean pulmonary artery pressure) presented in the passage.

(Choice D) Breathing frequency can be modulated voluntarily and cannot be combined with the data in the passage to draw accurate conclusions about pulmonary vascular resistance.

Educational objective:

Obesity-related changes in vascular resistance are accompanied by changes in mean arterial blood pressure and/or cardiac output.

Time spent:0 sec
Subject:
Biology

QID:403783
System:
Circulation and Respiration

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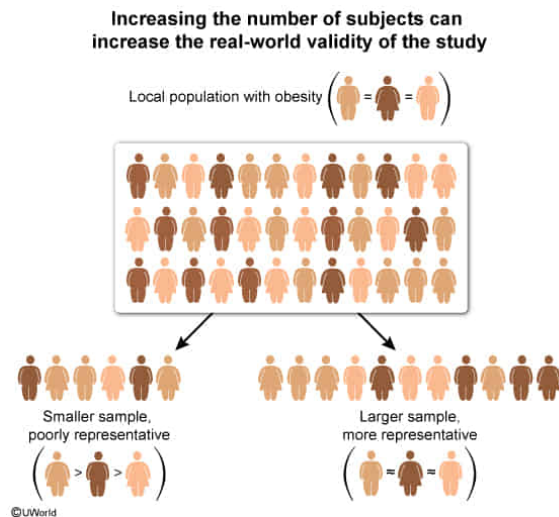
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In a subsequent clinical trial, researchers measured respiratory function in obese adults from the local population before and after these individuals were randomly assigned to either undergo a diet intervention or maintain their pre-study diet for 12 weeks. Which aspect of the experimental design would increase the real-world validity of the findings for the local population from which the sample was obtained?

- ☐ A. Restricting analyses to those individuals coming to weekly presentations on childhood obesity at the research center
- ☐ B. Nonrandomly selecting a sample of obese individuals above a certain height to ensure that such taller individuals are represented at a higher proportion in the study sample than in the local population [1%]
- ☒ C. Increasing the total number of subjects in the study to increase the likelihood of obtaining a representative sample [88%]
- ☐ D. Excluding individuals with peripheral obesity who did not adhere to their assigned diet on most days [9%]

Explanation:



Validity refers to the accuracy of research findings for a particular group or setting. Internal validity is the validity of a study within the sample studied; external validity is the validity of a study's findings as applied to a new sample.

Real-world validity is a type of external validity and refers to how well findings from a research study predict outcomes in a less-controlled setting and broader population outside the research study. For example, a study in which volunteers live in a metabolic laboratory and consume an experimental diet prepared by a dietician may be internally valid (ie, accurate within the context of the study) yet poorly predictive of outcomes in free-living individuals attempting to adhere to the diet themselves.

The question describes a study of respiratory function in obese individuals randomly assigned to one of two dietary regimens and asks for an experimental design aspect that would increase the real-world validity of the study findings for the local population from which the study sample was obtained. **Increasing** the total **number of subjects** in the study **would increase** the **likelihood** that **characteristics within the real-world population** of interest **are represented** in the study sample population. This would likely **increase** the **real-world validity** of the study for the local population.

(Choice A) Attendance at childhood obesity presentations is not necessarily related to representativeness with respect to the local obese population studied. Therefore, restricting analyses to attendees would *not* likely increase the real-world validity of the study for the local population.

(Choice B) Including taller members of the local obese population would increase the real-world validity of the study findings for the local obese population studied by the researchers. However, *nonrandomly overrepresenting* obese individuals above a certain height would introduce sampling bias that would likely *reduce* the real-world validity of the study findings for the entire local population of obese individuals.

(Choice D) The question does not indicate the proportions of individuals adhering to a particular diet or characterized as having peripheral or central obesity in the local

community. Therefore, excluding individuals with peripheral obesity who are nonadherent to their assigned study diet might bias the study and *reduce* the real-world validity of the study findings for the local population studied by the researchers.

Educational objective:

Validity refers to the accuracy of research findings, either within the study generating the findings or generalized beyond the context of the study; study design decisions influence validity.

Time spent:0 sec

Subject:
Biology

QID:403784

System:
Circulation and Respiration

Ventilatory actions, as well as nonventilatory actions necessary to expel particulate matter (eg, coughing), are achieved through the activation of inspiratory and expiratory muscles. The table shows levels of pleural and abdominal pressure under various conditions of air flow and intercostal muscle activation.

Muscle activation		Air flow	Pleural pressure	Abdominal pressure
external intercostals	internal intercostals			
-	-	-	-	+
+	-	+	--	++
-	+	+	-	+
-	+	-	+++	+++

Given the information in the table, are external and internal intercostal muscles inspiratory or expiratory muscles?

- ☐ A. External intercostals are inspiratory muscles, and internal intercostals are inspiratory muscles. [11%]
- ☐ B. External intercostals are expiratory muscles, and internal intercostals are expiratory muscles. [5%]
- ✓ ☒ C. External intercostals are inspiratory muscles, and internal intercostals are expiratory muscles. [62%]
- ☐ D. External intercostals are expiratory muscles, and internal intercostals are inspiratory muscles. [20%]

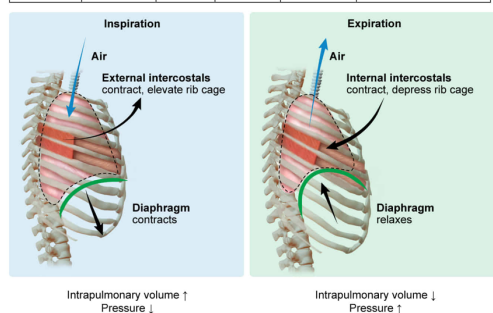
Correct

62% answered correctly

Explanation:

Activation of intercostal muscles during breathing in preparation for a cough

Muscle activation		Air flow	Pleural pressure	Abdominal pressure	Event
external intercostals	internal intercostals				
-	-	-	-	+	End of normal expiration
+	-	+	--	++	Inspiration
-	+	+	-	+	Expiration
-	+	-	+++	+++	Build-up of intrathoracic pressure before a (voluntary) cough



Coordinated contraction and relaxation of inspiratory and expiratory muscles are essential for supplying fresh air to the lungs during inspiration and removing gaseous waste from the lungs during expiration. In addition, contraction of expiratory muscles when the upper airway is closed following a deep breath generates high pressure inside the lungs. This increase in pressure is necessary for explosive actions (eg, coughing) that remove irritants (eg, inhaled particles or harmful gases) when the upper airway is opened and forceful expiration occurs.

During quiet breathing, inspiratory muscle contraction (primarily of the diaphragm) generates negative intrathoracic (ie, inside the thorax) pressure that expands the lungs. Pressure inside the **pleura** (the fluid-filled sac lining the lungs and the ribcage) becomes more negative during inspiration, which keeps the lungs stretched against the chest wall. Flattening of the diaphragm during inspiration also pushes the diaphragm against the abdominal contents, increasing pressure in the abdominal cavity. The elastic recoil of the stretched lungs and ribcage generates the positive pressure responsible for returning the lungs to their baseline volume during expiration (recall that the pressure and the volume of gases are **inversely related**).

With deeper or more vigorous breathing (eg, as with the deep inspiration and subsequent expiration of a voluntary cough), the external and internal intercostal muscles become more involved. The table data indicate that air can flow when either the external or the internal intercostals are activated.

The **negative intrathoracic pressure** observed when the **external intercostals contract** exerts a pulling effect on the lungs, contributing to lung expansion during **inspiration**. The **positive intrathoracic pressure** observed when the **internal intercostals contract** exerts a pushing effect on the lungs, causing the lungs to contract during expiration (or to build pressure inside if the airway is voluntarily closed). Therefore, the **external intercostals** are **inspiratory muscles**, and the **internal intercostals** are **expiratory muscles** (Choices A, B, and D).

Educational objective:

During breathing and protective actions involving inspiratory and expiratory muscles, activation of inspiratory muscles acts to reduce pressure inside the lungs and activation of expiratory muscles acts to increase pressure inside the lungs. The external intercostals are inspiratory muscles, and the internal intercostals are expiratory muscles.

Time spent:0 sec

Subject:
Biology

QID:403785

System:
Circulation and Respiration

The equation for calculating cardiac output (CO) using whole-body oxygen consumption (VO_2) and the systemic arterial-venous difference in O_2 concentration is shown.

$$\text{Cardiac output} = \frac{\text{VO}_2}{\text{Difference in arterial and venous O}_2}$$

What is the cardiac output of an individual whose tissues take up 350 mL of O_2 per minute and whose arterial-venous O_2 difference is 5 mL O_2 per 100 mL of blood?

- ☐ A. 0.7 mL/min [7%]
- ☐ B. 7 mL/min [6%]
- ☐ C. 70 mL/min [36%]
- ✓ ☒ D. 7000 mL/min [49%]

Correct

48% answered correctly

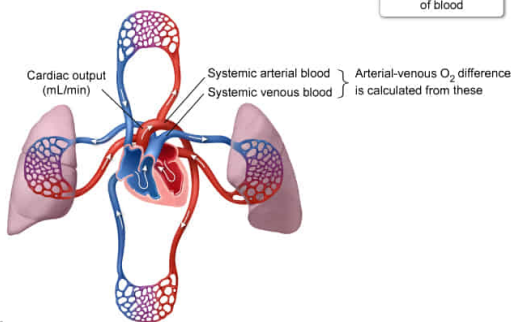
Explanation:

Calculation of cardiac output from whole-body oxygen consumption (VO_2)

Given equation and values:

$$\text{Cardiac output} = \frac{\text{VO}_2}{\text{Difference in arterial and venous O}_2}$$

350 mL O_2 per minute
5 mL O_2 per 100 mL of blood



The circulatory system **distributes blood** between the heart and the rest of the body's tissues. This distribution is achieved through the pumping action of the heart and the modulation of **blood vessel** diameter, which together create **pressure differences** that **regulate flow** along vascular circuits.

Typically, blood flow distribution is well matched to the needs of the tissues for delivery of fluid, oxygen, nutrients, and signaling molecules as well as for waste removal and **handling of metabolites**. The relative distribution of blood between the body's core and periphery also contributes to control of body temperature (ie, **thermoregulation**).

The total amount of blood pumped by the heart per unit time is called the cardiac output. Cardiac output can be calculated by numerous means, consistent with the heart's role as a pump and the circulatory system's function in delivery and removal of various molecules. One way of calculating cardiac output relates its pumping action to its function in systemic oxygen delivery, using the equation given in the question.

The question asks for the cardiac output calculated from whole-body oxygen consumption

(VO₂) and the difference in O₂ concentrations in the systemic arterial and venous blood. Using the equation provided along with the given oxygen uptake of 350 mL of O₂ per minute and the given arterial-venous O₂ difference of 5 mL O₂ per 100 mL of blood, the cardiac output can be calculated as follows:

$$\text{Cardiac output} = \frac{\text{VO}_2}{\text{Difference in arterial and venous O}_2}$$

$$\text{Cardiac output} = \frac{350 \text{ mL O}_2/\text{min}}{5 \text{ mL O}_2/100 \text{ mL blood}}$$

Simplifying and rearranging gives the following:

$$\text{Cardiac output} = \frac{350 \text{ mL O}_2/\text{min} \times 100 \text{ mL blood}}{5 \text{ mL O}_2/100 \text{ mL blood} \times \text{min}}$$

$$\text{Cardiac output} = \frac{70 \times 100 \text{ mL blood}}{\text{min}} = 7,000 \text{ mL blood/min}$$

Therefore, in an individual with the given whole-body O₂ uptake and systemic arterial-venous O₂ difference, the **cardiac output** is **7,000 mL blood/min**.

(Choices A, B, and C) These choices result from incorrect calculations of the cardiac output.

Educational objective:

The circulatory system circulates blood between the heart and the body's tissues and relies on the pumping action of the heart. This circulation delivers and removes molecules from the tissues and contributes to thermoregulation.

Time spent:0 sec

Subject:
Biology

QID:403786

System:
Circulation and Respiration

A commonly used approximation of mean arterial pressure (MAP) relates MAP to the measured systolic blood pressure (SBP) and diastolic blood pressure (DBP):

$MAP = \frac{2}{3}DBP + \frac{1}{3}SBP$. If the MAP increased by 10 mm Hg without a change in DBP, the new SBP would be:

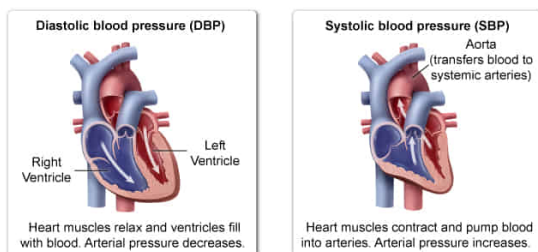
- ☐ A. the original SBP plus 10 [12%]
- ✓ ☒ B. the original SBP plus 30 [81%]
- ☐ C. the original SBP plus 60 [3%]
- ☐ D. the original SBP plus 90 [1%]

Correct

80% answered correctly

Explanation:

Change in systolic blood pressure responsible for the change in mean arterial pressure (MAP)



Original condition $MAP_{original} = \frac{2}{3}DBP_{original} + \frac{1}{3}SBP_{original}$

After change indicated in question $10mmHg + MAP_{original} = \frac{2}{3}DBP_{original} + \frac{1}{3}(SBP_{original} + \Delta SBP)$

↓ ignore unchanged terms
 $10mmHg = \frac{1}{3}\Delta SBP$
 ↓ Solve for ΔSBP
 $30mmHg = \Delta SBP$

If MAP increased by 10 mmHg without a change in DBP, the new SBP would be the original SBP plus 30 mmHg

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The driving force for the circulation of blood throughout the body is the pressure generated by the pumping action of the **heart**. This rhythmic pumping results in rhythmically changing (ie, "pulsatile") arterial flow and pressure. As blood flows through the arteries and capillaries, energy is lost, and the pulsatile nature of the blood flow and blood **pressure is decreased**. This results in more stable venous blood flow and much lower venous blood pressure compared to arterial blood flow and blood pressure.

Contraction of the left ventricle produces a peak in arterial blood pressure called the systolic blood pressure (SBP). During the time period between one contraction of the left ventricle and the following contraction, the arterial blood pressure declines, with the minimum blood pressure during this period defined as the diastolic blood pressure (DBP).

One way of describing blood pressure reduces it to a single mean value, the mean

arterial pressure (MAP). The question provides an equation often used to approximate MAP and asks for the new SBP that would result in an increase in MAP of 10 mm Hg, assuming DBP did not change. Using the equation given in the question and substituting (since it is given in the question that $\text{MAP}_{\text{new}} = \text{MAP}_{\text{original}} + 10 \text{ mm Hg}$ and $\text{DBP}_{\text{new}} = \text{DBP}_{\text{original}}$) gives

$$\text{MAP}_{\text{original}} + 10 \text{ mm Hg} = \frac{2}{3} \text{DBP}_{\text{original}} + \frac{1}{3} \text{SBP}_{\text{new}}$$

The **new SBP** can be expressed as the **original SBP plus** the amount SBP increased to achieve the new MAP (ΔSBP):

$$\text{MAP}_{\text{original}} + 10 \text{ mm Hg} = \frac{2}{3} \text{DBP}_{\text{original}} + \frac{1}{3} (\text{SBP}_{\text{original}} + \Delta \text{SBP})$$

Expanded, this becomes

$$\text{MAP}_{\text{original}} + 10 \text{ mm Hg} = \frac{2}{3} \text{DBP}_{\text{original}} + \frac{1}{3} \text{SBP}_{\text{original}} + \frac{1}{3} \Delta \text{SBP}$$

Compared with the equation given in the question, the expression to the left of the equals sign has increased by 10 mmHg and the expression to the right of the equals sign has changed by $\frac{1}{3} \Delta \text{SBP}$:

$$\text{MAP}_{\text{original}} + 10 \text{ mm Hg} = \frac{2}{3} \text{DBP}_{\text{original}} + \frac{1}{3} \text{SBP}_{\text{original}} + \frac{1}{3} \Delta \text{SBP}$$

Substituting from the original equation given in the question, the new equation becomes

$$\text{MAP}_{\text{original}} + 10 \text{ mm Hg} = \text{MAP}_{\text{original}} + \frac{1}{3} \Delta \text{SBP}$$

Subtracting $\text{MAP}_{\text{original}}$ from both sides and solving for ΔSBP gives

$$10 \text{ mm Hg} = \frac{1}{3} \Delta \text{SBP}$$

$$3 \times 10 \text{ mm Hg} = \frac{1}{3} \Delta \text{SBP} \times 3$$

$$30 \text{ mm Hg} = \Delta \text{SBP}$$

Therefore, the **new SBP** would be the **original SBP plus 30 mm Hg**.

(Choices A, C, and D) These answers represent *incorrect* calculations of the new SBP.

Educational objective:

Systolic blood pressure is the arterial pressure during ventricular contraction, and diastolic blood pressure is the arterial pressure during ventricular relaxation. Systolic

and diastolic blood pressure can be used to calculate mean arterial blood pressure.

Time spent:0 sec

Subject:

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